

Compiler Design

CSE - 420

Assignment 02

Summer 26

Note that:

1. *You will have to submit the soft copy*
2. *Make an cover page which contains you name, student ID and section*
3. *You do not have to submit the ungraded portion.*
4. *Submission form link: <https://forms.gle/JikdCdyaYhtLY1iL9>*

Deadline: 1 September,2026(11:59PM)

Question 1

Consider the following SDT.

Draw the Annotated Parse Tree and Evaluate it for the String: *if (a < b) c = f; else p = q * r ;*

```
int newlabel() {  
    return "L" || p;  
}
```

SDT:

P → {*p* = 0; *S.next* = newlabel();} **S** {*P.code* = *S.code* || label(*S.next*);}

S → {*p*++;} **Assign** {*S.code* = *Assign.code*;}

Assign → **id** = **E**; {*Assign.code* = *E.code* || gen(*id.lexeme* = *E.addr*);}

S → **if** ({*p*++; *B.true* = newlabel(); *B.false* = *S₁.next* = *S.next*; } **B**) {*p*++;}
 S₁ {*S.code* = *B.code* || label(*B.true*) || *S₁.code*;}

S → **if** ({*p*++; *B.true* = newlabel() || "_T"; *B.false* = newlabel() || "_F"; } **B**)
 {*p*++; *S₁.next* = *S.next*; } **S₁** **else** {*p*++; *S₂.next* = *S.next*; } **S₂**
 {*S.code* = *B.code* || label(*B.true*) || *S₁.code* || gen('goto' *S.next*) || label(*B.false*) || *S₂.code*;}

S → **while** ({*p*++; *begin* = newlabel(); *B.true* = newlabel() || "_T"; *B.false* = *S.next*; } **B**)
 {*p*++; *S₁.next* = *begin*; } **S₁**
 {*S.code* = label(*begin*) || *B.code* || label(*B.true*) || *S₁.code* || gen('goto' *begin*);}

S → {*p*++; *S₁.next* = newlabel();} **S₁** {*p*++; *S₂.next* = *S.next*;}
 S₂ {*S.code* = *S₁.code* || label(*S₁.next*) || *S₂.code*;}

B → {*p*++;} **E₁** **rel** {*p*++;} **E₂**
 {*B.code* = *E₁.code* || *E₂.code* || gen('if' *E₁.addr* *rel.op* *E₂.addr* 'goto' *B.true*); || gen('goto' *B.false*);}

E → {*p*++;} **E₁** **arith** {*p*++;} **E₂**
 {*E.addr* = newtemp(); *E.code* = *E₁.code* || *E₂.code* || gen(*E.addr* = *E₁.addr* *arith.op* *E₂.addr*);}

E → **id** {*E.addr* = *id.lexeme*; *E.code* = "";}}

E → **num** {*E.addr* = *num.value*; *E.code* = "";}}